Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/17/2019
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	05/21/2013	**End Date:**	10/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2020 per PI and S. Olson/GRC (Ed., 3/23/16) NOTE: Extended to 10/31/2015 per NSSC information (Ed., 3/24/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL10A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres, with the same principal investigator, Prof. Carlos Fernandez-Pello. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed. Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	During this reporting period on the MIST project most of the normal gravity work has concentrated in conducting experiments to study the effect of pressure and oxygen concentration on the downward spread of flames over the surface of PMMA rods of several diameters, and on the mass regression rate downstream of the flame. These experiments were conducted in the pressure vessel previously described. In addition, the experiments conducted in microgravity during the Burning and Suppression of Solids – II (BASS-II) tests in the ISS were compared with the normal gravity experiments to determine if conducting experiments at reduced pressure can be used to simulate conditions in microgravity. The results show that reductions of ambient pressure slow down the flame spread process approaching that observed at low gravity. The normal gravity and microgravity data are correlated in terms of a mixed convection parameter that describes the main controlling mechanisms of heat transferred. Although the correlation works well for the normal gravity data it does not work as well for the microgravity data. However, it provides information about what is to be expected in environments of variable ambient pressure, oxygen concentration, and reduced gravity, providing an insight for future designs when considering fire safety in spacecrafts.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Thomsen M, Fernandez-Pello C, Huang X, Olson S, Ferkul P. "The Effect of Buoyancy on Downward Flame Spread Over PMMA Cylinders." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
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Articles in Peer-reviewed Journals	Hernandez N, Reszka P, Fuentes A, Fernandez-Pello C. "Piloted ignition delay times on optically thin PMMA cylinders." Proceedings of the Combustion Institute. 2019;37(3):3993–4000. https://doi.org/10.1016/j.proci.2018.06.053 , Jan-2019
Articles in Peer-reviewed Journals	Xinyan H, Link S, Rodriguez A, Thomsen M, Olson S, Ferkul P, Fernandez-Pello C. "Transition from opposed flame spread to fuel regression and blow off: Effect of flow, atmosphere, and microgravity." Proceedings of the Combustion Institute. 2019;37(3):4117-26. https://doi.org/10.1016/j.proci.2018.06.022 , Jan-2019
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